Virtual Images

Exam code: 0625 & 0972

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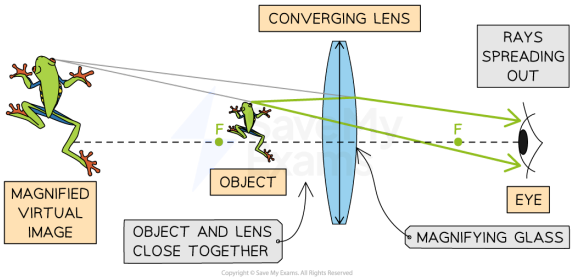
Converging lens – virtual image

Extended tier only

Constructing converging ray diagrams of virtual images

- A single lens placed at a distance **less than** the focal length of an object can be used as a **magnifying glass**

A converging lens ray diagram for an object placed less than f



Ray diagram showing light converging through a magnifying glass to form a magnified virtual image

- The image that forms will have the following properties:

The image forms…	at 2*f* (on the same side as the object)
The nature of the image is…	virtual
The orientation of the image is…	upright
The size of the image is…	magnified

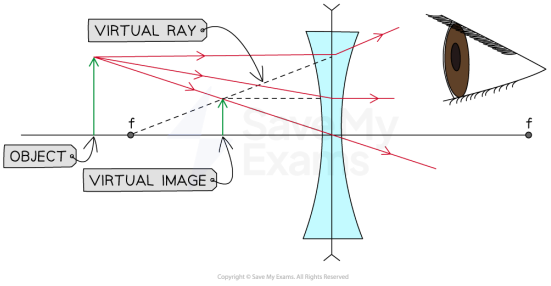
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

image

Image formation by a diverging lens

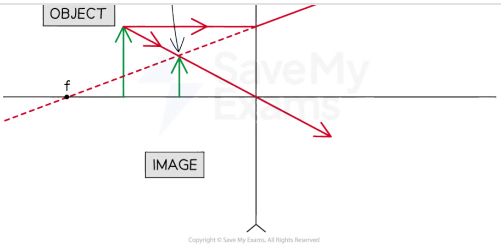
- No matter the position of the object all images formed by diverging lenses are:
  - **Virtual** (and not real)
  - **Upright** (the same as the object)
  - **Diminished** (smaller than the object)
  - On the **same side** of the lens as the object
- For an object placed at any distance away from the lens (further than the focal point or closer than):
  - The ray of light incident on the centre of the lens does not change direction
  - The rays of light parallel to the principal axis are refracted
  - These rays can be extrapolated backwards
    - It appears that they come from a **virtual focus**
  - A visible projection cannot be formed on a screen

Virtual image produced by a diverging lens with object beyond f



A diverging lens always produces a virtual image no matter the position of the object in relation to the focal point or the lens. Here the object is further away from the lens than the focal point.

Virtual image produced by a diverging lens with object closer than f



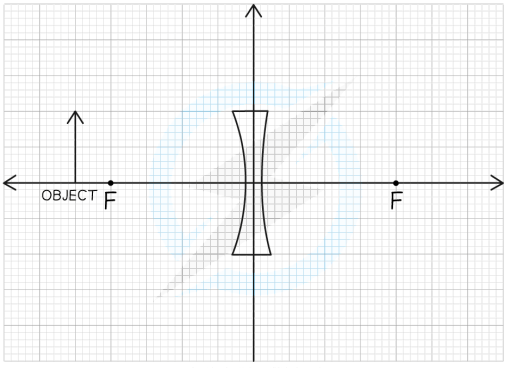
A diverging lens always produces a virtual image no matter the position of the object in relation to the focal point or the lens. Here the object is closer to the lens than the focal point.

Comparing converging & diverging lenses

- The image produced by a **converging** lens can be either **real** or **virtual**
  - This means the image can be inverted (real) or upright (virtual)
- The image produced by a **diverging** lens is always **virtual**
  - This means the image will always be upright

Worked Example

An object is placed outside the focal point of a diverging lens.



Complete the ray diagram by drawing where the image of this object will be seen.

Answer:

Step 1: Draw a line from the top of the object through the middle of the lens

- The top of the image lies somewhere along this line

Step 2: Draw a line from the focal point through the top of the lens

- The dashed line shows the continuation of the upward arrow
- The top of the image is where the two lines cross

Examiner Tips and Tricks

You are not expected to draw ray diagrams for a diverging lens but you must be able to explain how the virtual images are formed.

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